

Q. 1. With reference to the 'Basic Structure Doctrine', consider the following statements:

1. It was first formally introduced in the Golaknath case (1967).
2. The Constitution of India defines 'Basic Structure' under Article 368.
3. Judicial Review is considered a part of the Basic Structure.

Which of the statements given above is/are correct?

- A) 1 and 2 only
- B) 3 only
- C) 1 and 3 only
- D) 1, 2, and 3

Correct Answer: B (3 only)

Explanation: The Basic Structure doctrine was introduced in the Kesavananda Bharati case (1973), not Golaknath. The Constitution does not define Basic Structure; it is a judicial innovation. Statement 3 is correct as Judicial Review is a core pillar of the doctrine.

Q. 2. Which of the following conditions is/are essential for the formation of Karst topography?

1. Presence of soluble rocks like limestone or dolomite.
2. The rocks should be highly thin-bedded and poorly jointed.
3. The water table should be consistently high, touching the surface.

Select the correct answer:

- A) 1 only

B) 1 and 2 only

C) 2 and 3 only

D) 1, 2, and 3

Correct Answer: A (1 only)

Explanation: Karst requires soluble rocks and well-jointed (not poorly jointed) structures to allow water percolation. Crucially, the water table must be **low enough** to allow downward movement of water to create caverns and sinkholes.

Q. 3. The 'Drain of Wealth' theory, propounded by Dadabhai Naoroji, included which of the following components?

1. Salaries and pensions of British civil and military officials.
2. Interest on foreign debt incurred by the East India Company.
3. Profits of British merchants from free trade.

Select the correct answer:

- A) 1 and 2 only
- B) 2 and 3 only
- C) 1 and 3 only
- D) 1, 2, and 3

Correct Answer: D (1, 2, and 3)

Explanation: The "Drain" referred to the unilateral transfer of resources from India to Britain for which India got no proportionate economic return. This included "Home Charges" (salaries/pensions), interest on railway investments, and trade profits that were repatriated rather than reinvested in India.

Q. 4. If the Reserve Bank of India (RBI) decides to adopt an 'Expansionary' monetary policy, which of the following would it NOT do?

1. Cut and optimize the Statutory Liquidity Ratio (SLR).
2. Increase the Marginal Standing Facility (MSF) Rate.
3. Decrease the Bank Rate and Repo Rate.

Select the correct answer:

- A) 1 and 2
- B) 2 only
- C) 1 and 3
- D) 3 only

Correct Answer: B (2 only)

Explanation: An expansionary policy aims to increase liquidity. Increasing the MSF rate makes borrowing more expensive for banks, which is a **contractionary** (dear money) move. Reducing SLR, Repo, and Bank rates are all expansionary measures.

Q 5. In the context of 'Genome Editing', what is the primary role of the Cas9 protein in the CRISPR-Cas9 system?

- A) It acts as a "guide" to locate the specific DNA sequence.
- B) It acts as "molecular scissors" to cut the DNA at a specific location.
- C) It functions as a template for DNA repair.
- D) It acts as a ligase to join two DNA strands.

Correct Answer: B (Molecular scissors)

Explanation: In CRISPR-Cas9, the gRNA (guide RNA) identifies the target sequence, while the Cas9 protein performs the actual cleaving of the DNA strand.

Q 6. The 'Nagoya Protocol' is an international agreement related to:

- A) Protection of the Ozone Layer.
- B) Transboundary movement of hazardous wastes.
- C) Access to genetic resources and fair/equitable sharing of benefits (ABS).
- D) Reduction of greenhouse gas emissions.

Correct Answer: C (Access and Benefit Sharing)

Explanation: Supplementing the Convention on Biological Diversity (CBD), the Nagoya Protocol ensures that the benefits arising from the use of genetic resources are shared fairly with the provider country.

Q 7. The 'Panchayatan' style of temple architecture is characterized by:

- A) Five distinct entrance gateways (Gopurams).
- B) A main shrine surrounded by four subsidiary shrines.
- C) A temple built entirely out of five types of stones.
- D) A circular sanctum with five circumambulation paths.

Correct Answer: B (Main shrine with four subsidiary shrines)

Explanation: Seen frequently in Nagara style (e.g., Kandariya Mahadeva temple), the ground plan consists of a central large temple and four smaller temples at the corners, forming a "quincunx" pattern.

Q 8. Consider the following statements regarding 'Laterite Soils' in India:

1. They are formed under conditions of high temperature and heavy rainfall with alternate wet and dry periods.
2. They are rich in nitrogen, phosphate, and calcium.
3. They are widely used for making bricks for construction.

Which of the statements given above are correct?

- A) 1 and 2 only
- B) 2 and 3 only
- C) 1 and 3 only
- D) 1, 2, and 3

Correct Answer: C (1 and 3 only)

Explanation: Laterite soils are heavily leached, meaning nutrients like nitrogen and calcium are washed away, leaving them poor in organic matter. However, their hardened nature makes them excellent for brick-making.

Q.9 The 'Syadvada' doctrine, which suggests that all judgments are conditional and relative, belongs to which school of thought?

- A) Buddhism
- B) Jainism

C) Charvaka

D) Nyaya

Correct Answer: B (Jainism)

Explanation: Syadvada (the "theory of conditioned predication") is linked to Anekantavada. It asserts that reality is complex and can be viewed from multiple perspectives, none of which captures the absolute truth alone.

Q. 10 A person stands in the center of a rotating platform with their arms outstretched. If they suddenly pull their arms close to their body, what happens to their angular velocity?

- A) It decreases.
- B) It remains constant.
- C) It increases.
- D) It drops to zero.

Correct Answer: C (It increases)

Explanation: Based on the Law of Conservation of Angular Momentum, By pulling their arms in, the person decreases their Moment of Inertia . Since angular momentum must remain constant, the angular velocity must increase to compensate.